

Coding & Solution

Date	04 July 2026
Team ID	
Project Name	HDI Predictor Using Machine Learning
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/AP24110011041/HDI-Predictor-Using-Machine-Learning
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	• HDI prediction using Linear Regression • Data preprocessing and feature selection • Interactive Flask web application • Prediction of HDI score and category (Low, Medium, High, Very High) • User-friendly interface for input and results
Pending / Incomplete Features	• Graphical analytics dashboard • Multi-country comparison • User authentication and prediction history
Setup / Run Instructions	1. Install Python 3.10 and required libraries. 2. Run <code>pip install -r requirements.txt</code> (or install Flask, NumPy, Pandas, Matplotlib, Scikit-learn). 3. Navigate to the Flask folder. 4. Run <code>python app.py</code>. 5. Open <code>http://127.0.0.1:5000</code> in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The HDI Predictor uses a modular Flask architecture with a trained Linear Regression model (HDI.pkl). The code is organized, reusable, and maintained in a GitHub repository. It provides accurate HDI predictions through a simple and user-friendly web interface.